Content:

Classical versus Quantum Computation, bit, Superposition, Qubit and its properties, Dirac notation, Brief discussion on types of qubits. Neutral atom and ion qubit, Charge Qubit, The LC oscillator and its quantum analogue Quantum Harmonic oscillator, Equi-spaced energy levels, Need for anharmonicity.
Quantum Gates – Pauli Gates, Phase gate (S, T), Hadamard Gate, Two qubit gates – CNOT gate, Concept of Entanglement. Quantum Circuits, Numerical Problems.

The rapid advancement of information technology has significantly increased the demand for computational systems capable of solving increasingly complex problems. To overcome the limitations classical computers, researchers have explored a new computational paradigm known as quantum computation, which is based on the principles of quantum mechanics.

Classical computing and Quantum computing

Classical Computing	Quantum Computing
Conventional computing is based on the classical phenomenon of electrical circuits being in a single state at a given time, either on or off.	Quantum computing is based on the phenomenon of Quantum Mechanics, such as superposition and entanglement, the phenomenon where it is possible to be in more than one state at a time.
Information storage and manipulation are based on "bit", which is based on	Information storage and manipulation are based on Quantum Bit or "qubit", which is

voltage or charge; low is 0 and high is 1.	based on the spin of an electron or polarization of a single photon.
The circuit behaviour is governed by classical physics.	The circuit behaviour is governed by quantum physics or quantum mechanics.
Conventional computing use binary codes i.e. bits 0 or 1 to represent information.	Quantum computing use Qubits i.e. $\|0>$, $\|1>$ and the superposition state of both $\|0>$ and $\|1>$ to represent information.
CMOS transistors are the basic building blocks of conventional computers.	Superconducting Quantum Interference Devices or SQUID or Quantum Transistors are the basic building blocks of quantum computers.
In conventional computers, data processing is done in the Central Processing Unit or CPU, which consists of an Arithmetic and Logic Unit (ALU), processor registers and a control unit.	In quantum computers, data processing is done in a Quantum Processing Unit or QPU, which consists of several interconnected qubits.

Comparison of classical and quantum information

Quantum information	Classical information
Encoded to some property of a quantum system like photon polarization or spin of an electron.	Encoded to some property of a physical system obeying the laws of classical physics.
It is processed using quantum gates.	Processed using classical gates.
Fundamental unit of information is a qubit.	Fundamental unit of information is a bit.
It is difficult to store, transmit and process.	It is easy to store, transmit and process.
There is no way to copy unknown information.	It Easy to make copies of classical information.
Measurement of information destroys it.	Can be measured without disturbing it.

Bit: Bit (binary digit) is the smallest unit of information in classical computing and digital communication. It represents a single binary value, which can be either 0 or 1. All digital data (text, images, audio, video) is ultimately stored and processed as bits.

A Byte (8 bits) is commonly used to represent character, small numerical values etc.

The counterpart of a classical bit in quantum computing is Qubit. It's the basic unit in which of information in a quantum computer.

Qubit: It's the basic unit of information in a quantum computer.

A qubit, like a bit also makes use of two states $|0\rangle$ and $|1\rangle$ to hold information.

Mathematically, qubits |0> and |1> can be represented as column matrices:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}; \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

However, unlike classical bits, a qubit, |Ψ> can also be in a superposition state of $|0\rangle$ and $|1\rangle$ states.

It can be written as $|\Psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$
where α and β are generally complex numbers which represent the probability amplitudes of the states. When a qubit is measured, it only results in either $|0\rangle$ or $|1\rangle$.

Summation of probabilities

The probability of measuring the qubit in state $|0\rangle$ is $|\alpha|^2$, and the probability of measuring the qubit in state $|1\rangle$ is $|\beta|^2$. Since the total probability of observing all the states of the quantum system must add up to 100%, the modulus squared of the amplitudes must add up to 1. Thus, we have the following constraint:

$$|\alpha|^2 + |\beta|^2 = 1$$

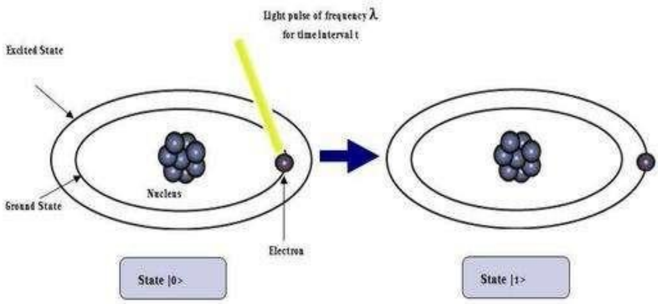
This is called Normalization rule.

Physical realization of qubits:

In a classical computer, the 0 and 1 bit mathematically represent the two allowed voltages across a wire in a classical circuit. Semiconductor devices called transistors are used to control what happens to these voltages.

What is a qubit made out of ?

1. Energy levels of an atom: Consider the electron in a hydrogen atom. It can be in its ground state (i.e. S orbital) or in an excited state. So we can also store a qubit of information in the quantum state of the electron, i.e., in the superposition of the Ground state $|0\rangle$ and the Excited state $|1\rangle$



2. Spin: Elementary particles like electrons and protons carry an intrinsic angular momentum called spin. Their spins can be used as qubits with $|0\rangle = |\uparrow\rangle$, $|1\rangle = |\downarrow\rangle$

3. Polarization of Photon: A linearly polarized photon can be either horizontally or vertically polarized with respect to some direction in which the photon is moving. Quantum researchers can create photons one at a time and encode qubits of information into their polarization.

Properties of Qubits:

1. A qubit can be in a superposed state of the two states 0 and 1.
2. Qubit measurements are inherently probabilistic due to superpositions

3. Owing to the quantum nature, the qubit collapses to any one of its state immediately when subjected to measurement. This means, one cannot copy information from qubits the way we do in the classical computers and is known as "no cloning principle".

Wave Function in Dirac notation:

A wave function (say ψ) represents the physical state of a system. According to Paul Dirac, the state of a system is described by a vector, called a state vector, in Hilbert space H. Depending on the degree of freedom (i.e. the type of state) of the system being considered, H may have infinite-dimensional.

Hilbert space H: It is a complex vector space. It has all the properties of linear vector space like vector addition and scalar multiplication. In addition, it satisfies inner product operation. An inner product is a generalization of the dot product It is a method of multiplying vectors together in a vector space, with the result being a scalar.

If ψ is a wavefunction, then in Dirac notation ψ is represented as $|\psi>$, which is called a **ket vector**.

Example: $\psi = A\, e^{-i\,k\,x}$

Dirac notation $|\psi\rangle = A\, e^{-i\,k\,x}$

Note: Only the notation of ψ is changed. The form of the wave function remains unchanged.

If ψ^* is the complex conjugate of ψ, then ψ^* is represented as $< \psi|$, which is called a **bra vector**.

Hence,

$$|\psi\rangle = A^* e^{i k x}$$

Note: Bra and Ket are derived from BRACKET. The notation was first derived by Paul Dirac.

Properties of KET and BRA vectors:

To every ket vector $|\psi\rangle$, there corresponds a bra vector $< \psi|$ and vice versa.

$$|\psi\rangle \leftrightarrow \langle\psi|$$

There is a one-to-one correspondence between ket vectors and bra vectors.

$a\ |\psi\rangle + b\ |\phi\rangle$ when complex conjugated becomes

$a^*\ |\psi\rangle + b^*\ \langle\phi|$

Where a and b are complex numbers.

Basis Vectors: In quantum mechanics, the "basis vectors" can be thought of as a set of mutually perpendicular vectors, one for each "dimension" of the space in which the state vector is expressed. The magnitude of a basis vector is one. There is a one-to-one correspondence between basis vectors and dimensions of the space.

Matrix form of wavefunction:

Consider a discrete and complete basis that is made up of an infinite set of kets $|\phi_1>$, $|\phi_2>$, $|\phi_3> \cdots$ etc.

The state vector $|\psi>$ can be written as a linear combination of kets $|\phi_1>$, $|\phi_2>$, $|\phi_3> \cdots$ etc as follows:

$$|\psi> = a_1|\phi_1> + a_2|\phi_2> + a_3|\phi_3> + \cdots + a_n|\phi_n> = \sum_{i=1}^{n} a_i|\phi_i$$

Where the coefficients $a_1, a_2, a_3 \ldots a_n$ represent the projection of $|\psi>$ onto $|\phi_i>$. Thus, a_i is the component of $|\psi>$ along the vector $|\phi_i>$.

Hence, $|\psi>$ can be represented as a **column vector (column matrix)** given by

$$|\psi> \rightarrow \begin{bmatrix} <\phi_1|\psi> \\ <\phi_2|\psi> \\ <\phi_3|\psi> \\ \vdots \\ <\phi_n|\psi> \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ . \\ . \\ a_n \end{bmatrix}$$

[A **column matrix** is a matrix having all its elements in a single column. The elements are arranged in a vertical manner. The order of a column matrix having n elements is n x 1]

The bra vector $<\psi|$ can be represented by a **row vector (row matrix)**:

$$<\psi| \rightarrow [\, <\psi|\phi_1> \;\; <\psi|\phi_2> \;\; <\psi|\phi_3> \;\; \ldots\ldots. \;\; <\psi|\phi_n> \,]$$
$$= [\, a_1^* \;\; a_2^* \;\; a_3^* \;\; \ldots\ldots.. a_n^*]$$

[A **row matrix** is a matrix having all its elements in a single row. The elements are arranged in a horizontal manner. The order of a row matrix having n elements is 1 x n]

Normalization:

A ket $|\psi>$ is normalized if $<\psi|\psi> = \sum_{i=1}^{n}|a_i|^2 = 1$

If $|\psi>$ is not normalized, we can multiply it by a constant 'a' so that

$< a^*\psi|a\psi > = |a|^2 < \psi|\psi > = 1$. This is the normalization equation.

The normalization constant a = 1/ $\sqrt{<\psi|\psi>}$

Inner Product:

If $\psi = \psi(x)$ and $\phi = \phi(x)$ are two wavefunctions, then their inner product can be defined as

$$(\psi, \phi) = \int \psi^* (x)\, \phi(x) dx$$

In Dirac notation (ψ, ϕ) is written as $< \psi|\phi >$.

Since the inner product (scalar product) is a complex number in quantum mechanics,

$$< \psi|\phi > = < \phi|\psi >^*$$

This property can be demonstrated as follows:

$$< \phi|\psi >^* = \left(\int \phi^* (x)\psi(x)dx \right)^* = \int \psi^* (x)\, \phi(x)dx = < \psi|\phi >$$

For any state vector $|\psi>$, $<\psi|\psi>$ is real and positive. If the state $|\psi>$ is normalized, $<\psi|\psi> = 1$

Therefore, $<\psi|\psi> = 0$ only if $|\psi> = 0$.

The inner product also called the dot product, signifies the projection of a wavefunction onto a particular state. This state could be some abstract spin state, a particular momentum state or even a specific value of position(x).

Thus $\psi(x)$ can be interpreted as the projection of $|\psi>$ onto the x basis

$|x>$.

Hence, we can write $c = <x|\psi>$, $\psi(p) = <p|\psi>$, $\psi(s) = <s|\psi>$

The above projections project an abstract wavefunction ψ onto real space $|x>$, momentum $|p>$ and spin $|s>$ respectively. This projected wave function is the probability amplitude of finding the system in the state (x,p,s), To get the probability we need to calculate the modulus squared of the amplitude. Thus

$$P(x) = |\psi(x)|^2 = |<x|\psi>|^2$$

Similarly $P(p) = |\psi(p)|^2 = |<p|\psi>|^2$

and $P(s) = |\psi(s)|^2 = |<s|\psi>|^2$

Condition for Orthogonality:

Two ket vectors $|\psi>$ and $|\phi>$, are said to be orthonormal if they are orthogonal and if eachof them is normalized.

$$\text{i.e.} <\psi|\phi> = 0, <\psi|\psi> = 1, \ <\phi|\phi> = 1$$

Matrix form of inner product:

$$\text{Let } |\psi> = a_1 |\phi_1> + a_2 |\phi_2> + a_3 |\phi_3> + \cdots + a_n |\phi_n>$$

$$= \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ . \\ . \\ a_n \end{bmatrix}$$

$$|\Phi> = b_1 |\phi_1> + b_2 |\phi_2> + b_3 |\phi_3> + \cdots + b_n |\phi_n>$$

$$= \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ . \\ . \\ b_n \end{bmatrix}$$

$$<\psi|\phi> = [\, a_1^* \quad a_2^* \quad a_3^* \quad \ldots\ldots\ldots a_n^*] \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ . \\ . \\ b_n \end{bmatrix}$$

$$<\psi|\phi> = a_1^* b_1 + a_2^* b_2 + a_3^* b_3 + \ldots\ldots\ldots a_n^* b_n$$

Operator:

An operator $\hat{A}$ is a Mathematical "Transformation" that when applied to a ket vector $|\psi>$ transforms it to another ket vector $|\phi>$ in the same Hilbert space and when it acts on a bra vector $<\psi|$ transforms it to another bra vector $<\phi|$ in the same Hilbert space.

Thus, we get: $\hat{A}\,|\psi> = |\phi>$ and $<\psi|\,A^{\dagger} = <\phi|$

As we will see if A = $A^{\dagger}$, the operator is Hermitian.

If $\hat{A}\,|\psi> = a\,|\psi>$, with 'a' real, then $|\psi>$ is an eigenfunction of $\hat{A}$ with eigenvalue 'a', where A has to be Hermitian.

Since, an operator transforms ket ψ (n×1 matrix) to another ket $|\phi>$(n×1 matrix), it is clear that the operator has to be n×n matrix.

Linear operators can be represented as square matrices in quantum mechanics.

Unity operator ($\hat{I}$) : It leaves any ket vector unchanged.

i.e. $\hat{I}\,|\psi> = |\psi>$

Identity matrix(I):

An identity matrix is a square matrix in which all the elements of principal diagonals are one, and all other elements are zeros. If any matrix is multiplied by the identity matrix, the result will be given a matrix.

$$I \equiv \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Example: The matrix form of ket vectors |0> and |1> can be written as:

$|0> = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1> = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, then

$$I\,|0> = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1x1 + 0x0 \\ 0x1 + 1x0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

I |1> = $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1x0 + 0x1 \\ 0x0 + 1x1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Transpose of a Matrix:

A^T denotes the transpose of a matrix A and it is obtained by swapping the rows with columns.

Suppose $A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$

$$A^T = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

Conjugate of a matrix:

A conjugate matrix is $\bar{A}$ a matrix obtained from a given matrix A by taking the complex conjugate of each element.

Hence, the conjugate of matrix A defined above is

$$\bar{A} = \begin{bmatrix} A_{11}^* & A_{12}^* & A_{13}^* \\ A_{21}^* & A_{22}^* & A_{23}^* \\ A_{31}^* & A_{32}^* & A_{33}^* \end{bmatrix}$$

Hermitian matrix:

A Hermitian matrix is a square matrix composed of complex numbers, and it is equal to its conjugate transpose.

Example: M = $\begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$

The conjugate transpose of the matrix is M^H or $M^\dagger = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$

Here, M = $M^\dagger$ Hence M is a Hermitian matrix.

Unitary matrix:

Unitary Matrix is a square matrix of complex numbers. The product of the conjugate transpose of a unitary matrix, with the unitary matrix itself, gives an identity matrix.

Example:

$U = \frac{1}{3}\begin{bmatrix} 2 & -2+i \\ 2+i & 2 \end{bmatrix}$

Taking conjugate $\overline{U} = \frac{1}{3}\begin{bmatrix} 2 & -2-i \\ 2-i & 2 \end{bmatrix}$

If we take the transpose of the above matrix, it is called a Hermitian matrix.

$$U^{\dagger} = \frac{1}{3}\begin{bmatrix} 2 & 2-i \\ -2-i & 2 \end{bmatrix}$$

$$\therefore\ U^{\dagger}.U = \frac{1}{3}\begin{bmatrix} 2 & 2-i \\ -2-i & 2 \end{bmatrix}\frac{1}{3}\begin{bmatrix} 2 & -2+i \\ 2+i & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Similarly, $U.U^{\dagger} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$

Hence, U is a unitary matrix.

Pauli matrices:

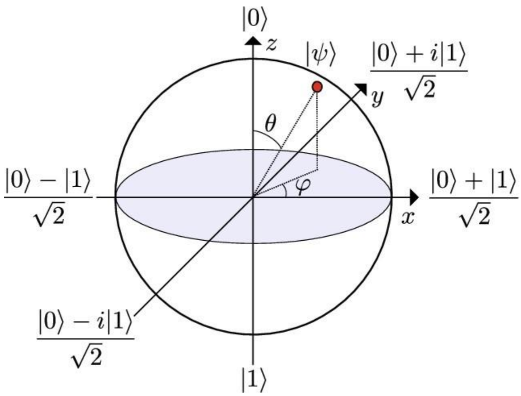
The Pauli matrices are a set of four 2x2 complex matrices. They are used to represent spin angular momentum. These matrices are Hermitian and Unitary. These matrices are very powerful in quantum computing as they can be used to represent quantum logic gates. They can set the rotational parameters for qubits. These matrices go by a variety of notations.

$\sigma_1 \equiv \sigma_x \equiv X \equiv \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $\qquad$ $\sigma_2 \equiv \sigma_y \equiv Y \equiv \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$

$$\sigma_3 \equiv \sigma_z \equiv Z \equiv \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Bloch sphere representation

Bloch sphere is a physical representation of all possible qubit states. It is a sphere of unit radius and the state of a qubit can be represented by a vector in this sphere. |0> is at the north pole, |1> is at the south pole, as shown in Figure below.



Using the spherical coordinate system, an arbitrary position of the state vector of a qubit canbe written in terms of the angles θ (elevation, the state vector makes from z-axis) and ϕ (azimuth, the angle of projection of the state vector in the x-y plane from the x-axis) it makes in the

$$|\psi> = e^{i\gamma}[\cos\left(\frac{\theta}{2}\right)\ |0>\ + e^{i\varphi}\sin\left(\frac{\theta}{2}\right)\ |1>]\ = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) \\ e^{i\varphi}\sin\left(\frac{\theta}{2}\right) \end{bmatrix}$$

Where γ is a Global phase and φ is the Local phase. The Global phase cancels out with its own complex conjugate and can therefore be set to zero, which is done, yielding the column matrix.

Note:

- For φ = 0 and $\theta = 0$, the state $|\psi>$ corresponds to |0> and is along z-axis.
- For φ = 0 and $\theta = 180^o$ the state $|\psi>$corresponds to |1> and is along -z-axis.
- When $\theta = 90^o$, $|\psi>$ is in the x-y plane.
- For φ = 90°, $|\psi> = \frac{1}{\sqrt{2}}\ (|0\rangle + i|1\rangle)$, is a superposition state along +y-axis.
- For φ = -90°, $|\psi> = \frac{1}{\sqrt{2}}\ (|0\rangle - i|1\rangle)$, is a superposition state along -y-axis.
- For φ = 0°, $|\psi> = \frac{1}{\sqrt{2}}\ (|0\rangle + |1\rangle)$, is a superposition state along the +x-axis.
- For φ = 180°, $|\psi> = \frac{1}{\sqrt{2}}\ (|0\rangle - |1\rangle)$, is a superposition state along the -x-axis.

Explanation of the Bloch sphere representation of a Qubit:

To understand the Bloch representation, we have to make sense of the projection of a spin operator onto any generalized axis. As we know that the projection of spin ½ onto any of the 3 axes, x,y,z gives us the components ħ/2 and – ħ/2. This result is true for projection about any generalized axis in 3 dimensions as well.

For a classical computer, the two logical states 0 and 1 are represented by the poles of a sphere.

In contrast, the state of a qubit can be represented by any point on the sphere. Since there are infinite points on the sphere, a qubit in principle has more capacity to store information compared to a classical bit.

Note: Bloch sphere represents the state of only one qubit. There is no generalization of Bloch sphere for multiple qubits.

Types of Qubits:
Neutral atom Qubit:

Atom which are not having electric charges are called Neutral atoms.

Each neutral atom represents a qubit by encoding information in two internal energy levels, often identified as the quantum $|0\rangle$ and $|1\rangle$ states.

The atoms are cooled to extremely low temperatures and held in place in a vacuum chamber using tightly focused laser beams, sometimes called optical tweezers.

To perform two-qubit gates, atoms are temporarily excited to Rydberg states (high-energy states with large electron orbits).

This creates strong, controllable interactions between neighbouring atoms—called the Rydberg blockade.

Lasers are used to detect fluorescence from atoms, revealing whether the qubit is in $|0\rangle$ or $|1\rangle$.

A major advantage of neutral-atom systems is that the atoms are identical by nature, which helps reduce noise and scaling complexities compared to manufactured qubits.”

Neutral-atom platforms can potentially scale to large arrays of qubits, because thousands of atoms can be trapped and controlled in a single system.

Trapped Ion Qubit:

A trapped-ion qubit is a quantum bit encoded in the internal electronic states of a single, laser-cooled ion that is held in place by electromagnetic fields in a vacuum chamber.

A neutral atom (commonly Ytterbium or Calcium) is used. One electron is removed using a laser forming a **positively charged ion**. A set of DC and RF electrodes creates an oscillating electric field that confines positively charged ions in a small region, typically arranging them as a linear chain. Keeps the charged ion suspended in vacuum. Prevents it from touching container walls.

The qubit is encoded in two long-lived electronic levels, such as hyperfine ground state and a metastable excited state.

$$|0\rangle = \text{Ground state}$$
$$|1\rangle = \text{Excited state}$$

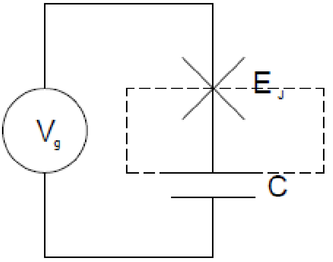
A precisely tuned laser pulse is applied. The laser causes transition between $|0\rangle$ and $|1\rangle$. By controlling pulse duration superposition is created which leads to single qubit operation

$$\alpha|0\rangle + \beta|1\rangle$$

Charge qubit:

The charge qubit, also known as Cooper pair box, consists of a small superconducting island coupled by a Josephson junction to a superconducting reservoir (the superconducting wire acts as the Cooper pair reservoir). The superconducting island is located between one of the plates of the capacitor and the insulating barrier of the Josephson junction is shown in figure. In this type of qubit, the charge is the variable controlled by the voltage source.

The number of cooper pairs



LC oscillator:

An LC oscillation occurs when a charged capacitor is connected to an inductor, causing the stored energy to transfer back and forth between electric and magnetic fields. The inductor and capacitor are connected in series, forming a closed loop with energy oscillating between the two components.

When the circuit is closed, the charged capacitor begins to discharge, transferring its energy into the magnetic field of the inductor as current flows. When the capacitor is fully discharged, the inductor's magnetic field holds the maximum energy. The inductor then maintains the current, recharging the capacitor with opposite polarity.

This cyclic process leads to continuous and sinusoidal oscillations of both charge on the capacitor and the current in the circuit.

Applying Kirchoff's Voltage law
$V_C+V_L = 0$
Where, $V_C = \frac{q}{C}$
And $V_L = L\frac{dI}{dt}$ with $I=-\frac{dq}{dt}$
The equation becomes

$$\frac{q}{C} + L\frac{d^2q}{dt^2} = 0$$

$$\frac{d^2q}{dt^2} + \frac{1}{LC}q = 0$$

This is the standard equation for simple harmonic motion, with charge oscillating sinusoidally at a natural angular frequency

$$\omega = \frac{1}{\sqrt{LC}}$$

Quantum Harmonic Oscillator:

The Quantum harmonic oscillator is the fundamental model in Quantum mechanics that describes a particle oscillates in a parabolic potential well of the form V(x) is $\frac{1}{2}m\omega^2x^2$. The motion of quantum harmonic oscillator never stops even at 0K temperature. Its energy states are discreate quantized energy levels given by

$E_n = (n+1/2)\hbar\omega$ (n=0,1,2,3----).

The ground state of the quantum harmonic oscillator is given $E_{gs}=E_0 = \frac{1}{2}(\hbar\omega)$ for n=0. The energy spectrum of quantum harmonic oscillator is equally spaced since the energy difference between any two successive energy levels is constant and is equal to ħω

$E_2-E_1 = E_3-E_2 = \ldots.\ E_{n+1} - E_n = \hbar\omega$

Converting a quantum harmonic iscillator into a qubit requires breaking the oscillator's equally spaced energy levels. Since the spacing between the energy levels are uniform, it is difficult to break the oscilaltor's equally spaced energy levels. Hence it is needed to introduce anharmonicity (unequal energy spacing) so that the transition from the ground state to the first excited state differs from the transition between any other two levels.
The anharmonicity can be achieved can be achieved in the parabolic potential by adding cubiv or quartic termd to the Hamiltonian. Hence the potential becomes

$$V(x) = \frac{1}{2}kx^2 + \lambda x^4$$

A positive λ steepens the potential well, squeezing higher energy levels further apart. A negative λ flattens the well, pulling higher energy levels closer together.

Bra Ket Representation of two qubit states

An un-normalized two Qubit state can be represented as $|\psi\rangle = |01\rangle$, where 0 is the state of the first qubit and 1 is the state of the second qubit. In general the outcome of the measurement of the state of the two Qubits will be independent of each other. However, as we have seen before if the two Qubits are entangled, the outcome of one Qubit instantaneously determines the outcome of measurement of the second Qubit. A maximally entangled state or Bell state has the property that the measured entanglement comes out to be a maxima. These states are used widely in **quantum communication, quantum teleportation,** and **Bell inequality tests**. An example of a maximally entangled state or Bell state is

$$|\psi\rangle=1/\sqrt{2}(|00\rangle+|11\rangle)$$

Quantum gates: Its action on Qubits:

Classical computer circuits consist of wires and logic gates. The wires carry information around the circuit, while the logic gates perform calculations and manipulate information, converting it from one to another. Computers manipulate bits using classical logic gates, such as OR, AND, NOT, NAND, etc.

Similarly, quantum computers manipulate qubits using quantum gates which are usually represented as matrices. A gate which acts on k qubits is represented by a 2k X 2k unitary matrix. The number of qubits in the input and output of the gate has to be equal. The action of the quantum gate is found by multiplying the matrix representing the gate with the vector which represents the quantum state. These are the rotation gates, which correspond to rotations about the x-, y-, and z- axes of the Bloch sphere. They are defined in terms of the Pauli gates, and so for convenience, we remind you now of the definitions of the Pauli gates:

Single Qubit gates

Pauli-X gate:

In classical computers, the NOT gate takes one input and reverses its value. For example, it changes the 0 bit to a 1 bit or changes a 1 bit to a 0 bit. It is like a light switch flipping a light from ON to OFF, or from OFF to ON.

Pauli-X gate is a quantum analogue of the classical NOT gate.

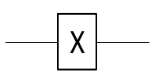
- The application of this gate rotates the qubit by 180° along the x-axis. It transforms $|0\rangle$ to $|1\rangle$ and vice versa.
- The matrix form of X-gate is obtained as follows

$$X = |0><1| + |1><0|$$

$$= \begin{bmatrix}1\\0\end{bmatrix} \begin{bmatrix}0 & 1\end{bmatrix} + \begin{bmatrix}0\\1\end{bmatrix} \begin{bmatrix}1 & 0\end{bmatrix}$$

$$= \begin{bmatrix}0 & 1\\0 & 0\end{bmatrix} + \begin{bmatrix}0 & 0\\1 & 0\end{bmatrix} = \begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$$

- Circuit representation



- Dirac notation
- $X\,|0\rangle = |1\rangle$ and $X\,|1\rangle = |0\rangle$
- When the qubit is in a superposition state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$, then

  $X\,|\psi\rangle = \alpha\,|1\rangle + \beta\,|0\rangle$
- In matrix form $|\psi\rangle = \begin{bmatrix}\alpha\\\beta\end{bmatrix}$

- Thus the action of X gate is :

$$X \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix}$$

Truth table

Input	Output
$\|0\rangle$	$\|1\rangle$
$\|1\rangle$	$\|0\rangle$
$\alpha\|0\rangle + \beta\|1\rangle$	$\alpha\|1\rangle + \beta\|0\rangle$

Pauli Y-gate

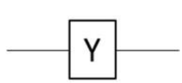
- The application of this gate rotates the qubit by 180^{o} along the y-axis.
- It transforms $|0\rangle$ to $i|1\rangle$ and $|1\rangle$ to $-i|0\rangle$.
- Matrix for of Y-gate

  $Y = i\,|1><0| - i|0><1|$

$$= i\begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} - i \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ i & 0 \end{bmatrix} - \begin{bmatrix} 0 & i \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

- Circuit representation



- Dirac notation

  $Y\,|0\rangle = i\,|1\rangle$

  $Y\,|1\rangle = -i\,|0\rangle$

Truth table

Input	Output
$\|0\rangle$	$i\|1\rangle$
$\|1\rangle$	$-i\|0\rangle$
$\alpha\,\|0\rangle + \beta\,\|1\rangle$	$\alpha\,\mathrm{i}\|1\rangle - \beta\, i\|0\rangle$

Pauli – Z gate:

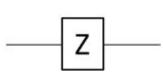
- The application of this gate rotates the qubit by 180° along the z-axis.
- It leaves $|0\rangle$ unchanged and flips the sign of $|1\rangle$ to $-|1\rangle$.
- Matrix form of Z-gate

  $$Z = |0><1| + |1> (-<1|)$$

  $$= \begin{bmatrix}1\\0\end{bmatrix}\begin{bmatrix}0 & 1\end{bmatrix} + \begin{bmatrix}0\\1\end{bmatrix}\begin{bmatrix}0 & -1\end{bmatrix}$$

  $$= \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix} + \begin{bmatrix}0 & 0\\0 & -1\end{bmatrix} = \begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}$$

- Circuit representation



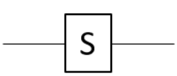
- Dirac notation:
  $Z|0\rangle = |0\rangle$ and $Z\,|1\rangle = -|1\rangle$

Truth table

Input	Output
$\|0\rangle$	$\|0\rangle$
$\|1\rangle$	-$\|1\rangle$
$\alpha\,\|0\rangle + \beta\,\|1\rangle$	$\alpha\,\|0\rangle - \beta\,\|1\rangle$

Phase gate (S gate)

- It rotates the qubit by $\frac{\pi}{2}$ radian along the z-axis.
- The effect of this gate is to modify the phase of the quantum state.
- It maps |0> $\leftrightarrow |0>$ and|1> $\leftrightarrow e^{\frac{i\pi}{2}}|1>$.
- Matrix representation S $\equiv \begin{bmatrix} 1 & 0 \\ 0 & e^{\frac{i\pi}{2}} \end{bmatrix}$
- Circuit representation



- Dirac notation

S $|0\rangle = |0\rangle$

S $|1\rangle = e^{\frac{i\pi}{2}}|1\rangle$

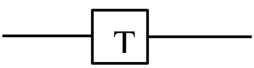
Truth table

Input	Output
$\|0\rangle$	$\|0\rangle$

$\|1\rangle$	$e^{\frac{i\pi}{2}}\|1\rangle$
$\alpha\|0\rangle + \beta\|1\rangle$	$\alpha\|0\rangle + \beta e^{\frac{i\pi}{2}}\|1\rangle$

T-gate

- Like S-gate, it modifies the phase of the quantum state.
- It maps $|0> \leftrightarrow |0>$ and $|1> \leftrightarrow e^{\frac{i\pi}{4}}|1>$.
- Matrix representation $T \equiv \begin{bmatrix} 1 & 0 \\ 0 & e^{\frac{i\pi}{4}} \end{bmatrix}$
- Circuit representation



- Dirac notation

  $T\,|0\rangle = |0\rangle$

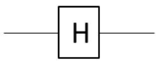
  $T\,|1\rangle = e^{\frac{i\pi}{4}}|1\rangle$

Truth table

Input	Output
$\|0\rangle$	$\|0\rangle$
$\|1\rangle$	$e^{\frac{i\pi}{4}}\|1\rangle$
$\alpha\|0\rangle + \beta\|1\rangle$	$\alpha\|0\rangle + \beta e^{\frac{i\pi}{4}}\|1\rangle$

Hadamard gate:

- It is one of the most important gates for quantum computing. If the qubit starts in a definite $|0\rangle$ or $|1\rangle$ state, the Hadamard gate puts each into a superposition of $|0\rangle$ and $|1\rangle$ states.
- Matrix representation $H \equiv \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
- Circuit representation



- Dirac notation

$$H\,|0\rangle = \frac{1}{\sqrt{2}}\,(\,|0\rangle + |1\rangle\,)$$

$$H\,|1\rangle = \frac{1}{\sqrt{2}}\,(\,|0\rangle - |1\rangle\,)$$

Truth table

Input	Output
$|0\rangle$	$\frac{1}{\sqrt{2}}\,(\,|0\rangle + |1\rangle\,)$
$|1\rangle$	$\frac{1}{\sqrt{2}}\,(\,|0\rangle - |1\rangle\,)$
$\alpha\,|0\rangle + \beta\,|1\rangle$	$\alpha\,\frac{1}{\sqrt{2}}\,(\,|0\rangle + |1\rangle\,) +$ $\beta\,\frac{1}{\sqrt{2}}\,(\,|0\rangle - |1\rangle\,)$

Two Qubits gate:

CNOT gate :

- It is a Controlled NOT (CNOT) gate.

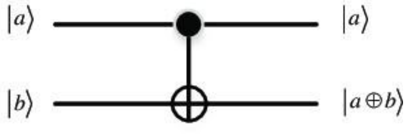
- It acts on two qubits.
- It performs the NOT operation on the second(target) qubit only when the first(control) qubit is |1>, otherwise leaves it unchanged. Diagrammatically, the solid dot represents the control Qubit and the circle with plus sign inside represents the Target Qubit.

Truth table of CNOT gate:

Input		Output	
a	b	a	a⊕b
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

Matrix and symbolic representation of the CNOT gate:

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$



Here $|a\rangle$ represents control qubit and $|b\rangle$ represents target qubit.

Dirac notation of CNOT Gate :

CNOT $|00\rangle = |00\rangle$; CNOT $|01\rangle = |01\rangle$

CNOT $|10\rangle = |11\rangle$; CNOT $|11\rangle = |10\rangle$

Quantum Entanglement:

Quantum Entanglement refers to two or more quantum systems which are linked in such a way that the measured state of one system depends on the state of the other, even if the two systems (electrons or photons) are far apart, like one is in Earth and the other on the Moon.

The question arises how can this be possible, because it apparently contradicts the fundamental axiom of Physics, namely nothing can travel faster than light. However a careful consideration reveals that there is no violation of Special Theory of Relativity. The reasons are :

1. No Physical Information Travels Faster than Light when measurement is made on entangled particles.

Entanglement involves correlations, not communication. STR forbids faster-than-light information transfer, not instantaneous correlations. Although measurement of one entangled particle seems to *instantly* determine the other's state, no usable information is transmitted between the particles. We cannot control the outcome of a quantum measurement to send a message.

2. Measurement Outcomes Are Random

When you measure one entangled particle, its result is random (e.g., spin-up or spin-down). While the other particle's result is correlated, you cannot predict or control the first result, so you can't use it to signal anything.

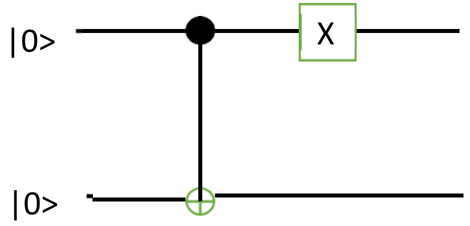
3. Causality Is Preserved

Because no information is sent, causality (i.e., cause happens before effect in all reference frames) is maintained. Even if entanglement correlations appear "instantaneous," no cause-and-effect chain moves faster than light.

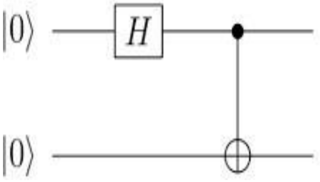
4. Relativity and Quantum Mechanics Coexist. Relativistic quantum field theories (Quantum Electrodynamics) respect both quantum mechanics and STR. Entanglement is fully compatible with the mathematical framework of STR.

Quantum circuits:

A quantum circuit is required to carry out computations on a quantum computer. It consists ofa series of operations referred to as quantum gates. These quantum gates, which are assignedto certain qubits, change the quantum states of some of the qubits, causing those qubits to perform the calculations required to solve a problem.



CNOT gate leaves the target qubit unchanged as the control qubit is |0>. In the second step, X-gate flips the control qubit to |1>. Hence the output is |10>.



The states change from the start to the end after every gate:

The Hadamard gate changes |0> to $1/\sqrt{2}$ ($|0\rangle$ +$|1\rangle$).

The input of the CNOT gate is the two qubit state $1/\sqrt{2}(|00\rangle+|10\rangle)$. CNOT acting on $|00\rangle$ generates $|01\rangle$ while CNOT acting on $|10\rangle$ generates $|11\rangle$.

Hence CNOT changes the output to $1/\sqrt{2}$ ($|00\rangle$ + $|11\rangle$). This is a maximally entangled 2 Qubit state as we have seen before. The above example illustrates how we can generate a maximally entangled 2 Qubit state using Hadamard and CNOT Gates.